Mudit Arora

San Francisco Bay Area | +1 (602) 545-7387 | muditarora31@gmail.com | aroramudit | Mudit-Arora | Portfolio

EDUCATION

Master of Science, Artificial Intelligence — Dec 2025
University of California, Santa Cruz — Santa Cruz, CA
- **Courses**: Machine Learning, Deep Learning, Natural Language Processing, LLM Agents, Generative AI, Conversation AI, Multimodal AI, Data Science, AI Agents, AI in Games — GPA: 3.86

Bachelor of Science, Computer Science — May 2024
Arizona State University — Tempe, AZ
- **Awards/Honors:** Magna Cum Laude, Dean's List, New American University Scholar, SUN Award — GPA: 3.77
- **Courses**: Artificial Intelligence, Data Structures & Algorithms, Software Engineering, Computational Biology

SKILLS

Programming Languages: Python, R Programming, C++, MATLAB, Java, JavaScript, TypeScript, Swift
Frameworks/Libraries: PyTorch, TensorFlow, scikit-learn, Keras, MLX, NLTK, Pandas, NumPy, LangChain, AutoGen, matplotlib, OpenCV, HTML/CSS, React, Next.js, Node.js
Tools/Technologies: GCP, Azure, Git, MySQL, Docker, AWS (S3), Postman, Hugging face, Ollama, Make.com
Domain: Artificial Intelligence, Machine Learning, Deep Learning, Data Science, NLP, LLM, RAG, GenAI, AI Agents, Computer Vision
Certifications: Technical Interview Prep (CodePath), Web Development (CodePath)

WORK EXPERIENCE

AI Engineer — Feb 2026 – March 2026
EzMedTech AI — Remote
- Built Voice AI agents for clinical appointments using **ElevenLabs** using propriety & open-source LLMs.

Deep Learning Researcher — May 2025 – Feb 2026
Uniphore — Palo Alto, CA
- Generated **500+ synthetic conversations** based on frameworks like ReAct & Pre-Act on multiple domains such as Banking, Travel, etc. for a flexible dialog agent system that is capable of multi-turn reasoning and dynamic tool calling on normal, happy & unhappy paths.
- Trained a series of **Qwen Small Language Model** using **Supervised Fine Tuning (SFT**) to generalize tool use and tuned **Reinforcement Learning with Verifiable Rewards (RLVR)** on Action Reward, Tool Reward, and Semantic Reward.
- Achieved **50% and 66.7% success rate** compared to base models (**12x improvement**) on seen domain & unseen domain respectively.

AI Software Engineer Intern — May 2025 – Sept 2025
CRED — San Francisco, CA
- Worked on internal automation tools to help PMs in taking notes, reviewing transcripts, identifying key problems, solutions, and improvements **saving upto 2hrs.**
- Designed an AI bug fixing agent for writing fixes, Q&A, and reviewing the bugs helping the developers **saving upto 4hrs**.
- Crafted features for CRED's Chrome Extension utilizing LLMs for intelligent preprocessing and semantic chunking to scrape webpage data, **achieving 84% accuracy** in the benchmark test, and optimized the process to retrieve real-time data to users efficiently.
- Optimized CRED's AI Slack Bot for better workflow, **reducing the wait time by 15%** by using OCR models.

Machine Learning Researcher — Aug 2023 – May 2024
Mayo Clinic — Tempe, AZ
- Fine-tuned **Google's T5**-based LLM using **Python**, **PyTorch**, and **scikit-learn**, achieving **87% accuracy** in extracting social determinants of health from clinical notes and predicting patient readmission within 30 days.
- Assisted hospitals in **reducing admission rates**, leading to **cost savings** in patient care management and improved clinical decisions.

Software Quality Assurance Intern — May 2022 – Aug 2022
Knight Transportation — Phoenix, AZ
- Innovatively constructed an agile approach, authoring detailed test cases and effectively resolving critical bugs via **Microsoft Azure DevOps** and **Elasticsearch**; optimized development processes, resulting in a **28% reduction in bug resolution** time.
- Orchestrated a collaborative effort with Backend Engineers to optimize User Experience, resulting in a **42% increase in app engagement** and a **23% decrease in user complaints**.

PROJECTS

Slug Meditate – CruzHacks 2025 Winner
- Built a VR meditation web app pipeline, utilizing **Google's Gemini API** to transform user text prompts into AI generated image (**Imagen 3**) and video (**Veo 2**) that then transforms it into a 3D scene mapping (**Gaussian Platting**), then add an AI generated music (**MusicFX**) that compliments the meditative vibe, and finally rendering the VR immersion (**Niantic Studio by 8th Wall**).
- Achieved a **success rate of 87%** in rendering immersive by processing over **14 unique** user prompts.

Blindspot – Smallest.ai Hackathon Runner Up
- Engineered an accessibility-focused iOS app for blind/visually impaired users featuring on-device ML-powered object/scene detection (**CoreML + VisionKit)** and voice AI (**Smallest.ai + Groq**); delivered real-time performance with **<2s response latency**, reducing navigation uncertainty; built using **Swift/SwiftUI** and **Python (FastAPI)**.

EduMUSE
- Developed an AI-powered tutoring system using **CrewAI** multi-agent architecture that transforms PDF study materials into personalized learning experiences with automated summaries, quiz generation, and podcast-style audio content using **GPT-4o**, **SerperDev**, and **ElevenLabs**, reducing study material **processing time by 68%.**